# Ahmed Ayman at ImageEval 2026 Shared Tasks: Measurement-First Systems for Spoken VQA and Cultural Accuracy Evaluation

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## Abstract

We describe our systems for both ImageEval 2026 shared tasks. On Subtask 1a (spoken visual QA) we reach 0.962 on English and 0.834 on Modern Standard Arabic against baselines of 0.664 and 0.398, ranking first on the English track and second on MSA, while on Task 2 (CRAI-Bench) our judge ensemble reaches a test Spearman of 0.673 against 0.533 for our reproduction of the organizer baseline. Three effects account for most of these margins: answer-token bias (digit labels suppress the answer 0, and ordinal words are worth 0.35 accuracy), Arabic speech recognition (isolated from visual reasoning by the parallel track design), and tie-breaking (calibration collapses ranks and a second judge separates them, while error correlation of 0.89 rules out cancellation). A post-submission study on degraded audio finds four transcript-level interventions null and measures a 9% answer-flip rate between identical runs at temperature 0, a noise floor for single-run ablations.

**Keywords:** spoken visual question answering, cultural evaluation, Arabic NLP, LLM-as-a-judge, speech recognition, selection bias, ensemble calibration

## 1 Introduction

Both ImageEval tasks (Abdaljalil et al., 2026) test whether systems handle culturally grounded Arabic multimodal content, a setting where models measurably prefer Western entities (Naous et al., 2024). Subtask 1a asks which of three spoken options an image supports, in parallel English and MSA tracks over the same images, while Task 2 asks a system to score AI-generated images of Qatari cultural scenes the way human annotators from the region do.

We participated in both language tracks of Subtask 1a and in Task 2, and the paper is organized around one question: where do the margins actually come from? Each design choice follows from a quantitative observation, and each headline number is traced to the effect that produces it. Appendix A specifies both systems to re-implementation level.

## 2 Tasks and data

In Subtask 1a, each item pairs an image with a recording of a spoken question followed by three spoken options, with nothing given as text, and systems output the index of the correct option, ranked by accuracy. The images cover culturally specific content from 18 Arab countries, each item tagged with country and cultural category, and the data builds on the OASIS spoken VQA collection (Alam et al., 2025) and its hallucination analyses (Mousi et al., 2026b,a). Training, dev and devtest speech is synthesized by voice cloning, whereas the blind test speech is recorded by human speakers.

In Task 2, each instance is a triple: an authentic reference photograph of a Qatari cultural scene, a caption, and an image generated from that caption. Systems predict five scores in $[0, 1]$: cultural element accuracy (CEA), contextual coherence (CC), cultural specificity (CS, restricted to the tiers $0, 0.25, \ldots, 1$), cultural integrity (CI), and a hallucination penalty (HP), combined as $0.3\,\mathrm{CEA} + 0.2\,\mathrm{CC} + 0.2\,\mathrm{CS} + 0.2\,\mathrm{CI} - 0.1\,\mathrm{HP}$. Ranking is by Spearman correlation between predicted and human composite scores, with MAE as tiebreak. Each split pairs its reference scenes with five captions per scene at decreasing cultural specificity, from fully Qatari-specific (v1) down to entirely generic (v5), and this specificity ladder is what defeats caption-anchored element checking (Section 4.2). Table 1 summarizes both datasets.



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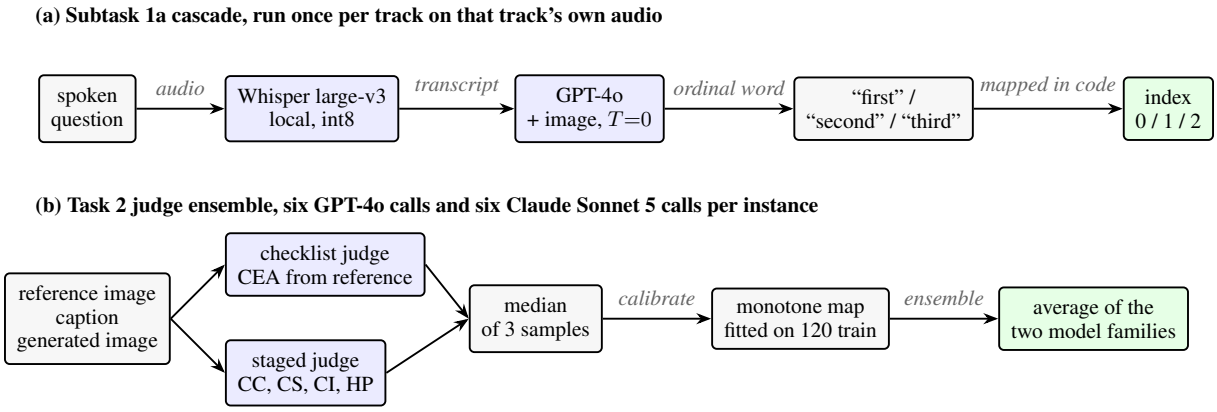



Figure 1: Both submitted systems share two rules: the model must write its evidence before its decision (the cascade reconstructs the question and options first, and the judges record element-by-element verdicts before any score), and the final answer format is produced by code, never by the model.

| **Subtask 1a** (per track) | EN | MSA |
|---|---|---|
| train / dev / test items | 3,000 / 500 / 1,000 | |
| dev items used here | 100 | 100 |
| dev audio (synthetic), mean s | 15.6 | 18.0 |
| test audio (human), mean s | 11.4 | 8.7 |
| dev gold labels 0/1/2, % | 38 / 35 / 27 | |
| test labels 0/1/2, % (est.) | ∼55 / 42 / 3 | |
| **Task 2** | | |
| instances train / dev / test | 120 / 40 / 40 | |
| reference scenes | 8 per split, disjoint | |
| human median composite | 0.48 | |
| human HP = 0 | 116 of 120 train | |

Table 1: Data statistics. The 1a test label split is estimated from our 0.962-accurate English predictions.

| | Dev | | Test | |
|---|---|---|---|---|
| System | EN | MSA | EN | MSA |
| Baseline (published) | – | – | 0.664 | 0.398 |
| Joint, digit labels | 0.560 | – | – | – |
| Joint, ordinal labels | 0.910 | 0.410 | 0.928 | – |
| Whisper → Qwen | – | 0.590 | – | – |
| Whisper → GPT-4o | **0.970** | **0.950** | **0.962** | **0.834** |

Table 2: Subtask 1a accuracy. Dev is 100 items per track, and test is the official score on 1,000 blind items. The submitted cascade ranked first (EN) and second (MSA) among participating teams.

# 3 Subtask 1a: Spoken VQA

## 3.1 Systems

We compared two designs per track. The **joint** system gives Qwen2.5-Omni-3B (Qwen Team, 2025) the image and audio in one request, 4-bit on one consumer GPU. We chose this model because it is the organizers' official baseline for the subtask, which makes our joint results directly comparable to the published baseline scores, and because its 3B size keeps the fully local variant practical. The **cascade** (Figure 1a) transcribes with Whisper large-v3 (Radford et al., 2023) locally, then GPT-4o (OpenAI, 2023) receives the image and the transcript, with a warning that the transcript is automatic and must be reconstructed first. The cascade is the submitted system on both tracks, and the joint system is the end-to-end baseline and diagnostic instrument (Section 3.3).

## 3.2 Effect of answer labeling

The largest single effect on the English track comes from the answer format rather than from perception. With the digit labels of the organizers' baseline prompt, the joint model scores 0.56 on English dev while its written reasoning describes the images correctly. The failure is concentrated in one class: Figure 2a shows first-option recall at 3/38 under digit labels against 36/38 when the same options are called first, second and third. A digits prompt asserting all options equally likely reached 0.86, so the effect is a token preference, not a belief about the distribution. This matches the option-identifier bias of Zheng et al. (2024) and the order sensitivity of Pezeshkpour and Hruschka (2024). To our knowledge, the specific finding that 0 is the token models resist has not previously been reported for a spoken setting.

## 3.3 Diagnosing the MSA gap with the parallel tracks

Ordinal labeling lifts English to 0.91 yet leaves MSA at 0.41, so a different problem must re-

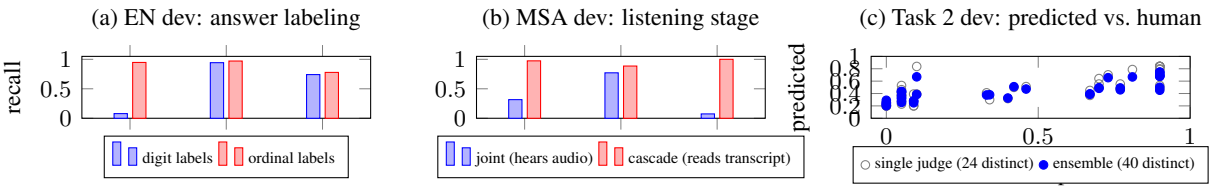



Figure 2: Error analysis on real predictions. (a) Renaming the digit labels 0/1/2 as ordinal words repairs first-option recall from 3/38 to 36/38. (b) The MSA failure is listening, not reasoning: the same 3B model recovers third options at 2/27 from raw Arabic audio, the cascade at 27/27 from a transcript. English needs no such repair, since the same substitution moves it only from 0.91 to 0.97 (Table 2). (c) All 40 Task 2 dev instances. The calibrated single judge (grey) collapses onto horizontal bands of ties (seven instances share $y$=0.837), and tied predictions take the average rank, capping Spearman. The average (blue) separates the bands.

main on the Arabic side. Because the two tracks share their images and answers, a visual or cultural deficit is ruled out, and the only input that differs between them is the audio. Figure 2b gives the per-class picture: on MSA the joint model recovers third options at 2/27, and its logs never reproduce the Arabic options, in one case answering "I'm not sure which option you're referring to." The deficit can be separated into its two stages. Giving the same 3B model a Whisper transcript instead of raw audio, so that Qwen still performs the reasoning but no longer the listening, lifts MSA from 0.41 to 0.59. Swapping the reasoning model as well, from Qwen to GPT-4o, gives the full cascade and lifts MSA to 0.95 with third-option recall 27/27. Both stages were therefore limiting: of the 0.54 gap between the joint system and the full cascade, the listening substitution recovers 0.18 and the reasoning substitution the remaining 0.36, a one-to-two split.

### 3.4 Development-to-test shift

The move from synthetic development speech to human test speech affects the two tracks very differently. English accuracy barely changes (0.970 to 0.962), while MSA falls from 0.950 to 0.834. Since both tracks run the same architecture, the fragile component must be Whisper's Arabic recognition under real speakers, which Section 5 examines in detail.

This shift can also mislead error analysis. Our English test predictions choose the third option only 17 times in 1,000, a pattern that resembles the MSA failure, yet they score 0.928: the skew simply mirrors the gold label distribution (Table 1), so a skewed prediction distribution is not by itself evidence of a broken system.

## 4 Task 2: CRAI-Bench

We treat Task 2 as an alignment problem: a generous LLM judge must be made to score the way a panel of strict human annotators does. We therefore started from the released annotations rather than from prompt ideas.

### 4.1 Design constraints from the annotations

Three statistics on the 120 training instances shaped the design (Table 1): first, the annotators are strict (median composite 0.48, a quarter below 0.15) while the organizer baseline predicts between 0.85 and 1.0. Second, mean cultural specificity is 0.31. Third, HP is 0.0 on 116 of 120 instances, so HP correlation is mostly noise and the right objective is low MAE. These facts appear verbatim in our prompts as calibration anchors.

### 4.2 System components

A **staged judge** (GPT-4o) must describe the generated image, verify every expected element with one line of visual evidence, and list fabricated elements before it may score CC, CS, CI and HP. In brief, CC asks whether the depicted elements combine into a culturally plausible scene, CS rates how specifically the generated image alone reads as Qatari on the five tiers, CI rates truthfulness and respect (stereotyping or costume-like inaccuracy lowers it), and HP may only penalize fabrications the judge has named in its list. Each score is defined in terms of its recorded verdicts, following the form-filling logic of G-Eval (Liu et al., 2023) with the stages made mandatory. In code, HP is forced to 0 whenever the fabrication list is empty, because a guessed penalty is far more likely to be noise than signal, and the composite weights that noise at $-0.1$. Our HP MAE is 0.035 against the baseline's 0.338.

| System | Dev | | Test |
|---|---|---|---|
| | $\rho$ | MAE | $\rho$ |
| Organizer baseline (reproduced) | 0.643 | 0.224 | – |
| Staged judge alone | 0.644 | 0.276 | – |
| + baseline CEA | 0.745 | 0.217 | 0.533 |
| + checklist CEA | 0.799 | **0.199** | 0.579 |
| + calibration | 0.813 | 0.204 | 0.599 |
| Same pipeline on Sonnet-5 | 0.805 | 0.263 | – |
| average of the two | **0.855** | 0.232 | **0.673** |

Table 3: Task 2 composite Spearman and MAE. The four test entries are our four submissions, in order, and they rank exactly as their dev scores do.

A separate **checklist judge** scores element accuracy (CEA). The natural design would anchor its checklist in the caption, but that scores only 0.256 against the baseline's 0.616: on culturally generic captions the expected-element list is empty, and the judge emits a vacuous 1.0 where the human score is 0.0, because the annotators score against the reference scene rather than the caption. The checklist judge therefore lists 4–8 concrete elements visible in the reference image and verifies each in the generated image, TIFA-style (Hu et al., 2023) but anchored in the image rather than the text.

Scores are medians of three samples at temperature 0.4 (Wang et al., 2023), corrected by an isotonic (monotone) map fitted on the training annotations, which fixes the judges' generous scale without changing the rank order within any dimension. Finally the whole pipeline runs again on Claude Sonnet 5 and the two score tables are averaged with equal, unfitted weights, since weights chosen on 40 dev instances would overfit them. A post-hoc sweep of the mixing weight supports this: every mixture between 0.3 and 0.7 outperforms both single judges on dev, and the equal-weight score of 0.855 sits within 0.003 of the best point in the sweep, so the ensemble gain does not depend on tuning the weight.

### 4.3 Analysis of the ensemble gain

The average is the largest single test gain (+0.074). Error cancellation, the usual justification for ensembles, does not explain it: the judges' composite errors correlate at 0.89, the average is closer to gold than both judges on only 4 of 40 dev instances, and MAE worsens (0.204 to 0.232). A diverse-panel account (Verga et al., 2024) does not fit either: the second judge alone is no better than the first (0.805 vs. 0.813).

Figure 2c points to the actual mechanism: the metric is a rank correlation, and calibration creates ties. The isotonic map collapses 40 dev predictions onto 24 distinct values (27 on test), and tied predictions take the average rank, capping correlation against a gold ranking with almost no ties. The second judge restores resolution (40 distinct on dev, 39 on test).

Of the 19 dev pairs the single judge scored identically that differ in gold, the average separates all 19 and orders 15 correctly. Calibration improves the ordering while destroying resolution, and the ensemble restores it. It also softens without fixing the single judge's largest error, a human-0.10 instance scored 0.837 alone and 0.67 averaged (top left of Figure 2c). Under an MAE ranking we would not expect the second judge to be worth its doubled inference cost.

## 5 Post-submission robustness analysis

One result above still lacks an explanation: the MSA drop from 0.950 to 0.834. This section pursues it post-submission, with the official results unchanged in Tables 2 and 3. The dev split cannot probe the failure, its synthetic audio saturating at 0.95, so we built a surrogate: the 100 labeled dev clips degraded with room echo, telephone-band filtering, pink noise at 12 dB SNR and low-bitrate compression, parameters fixed before any run. Under the submitted system this set scores 0.840 against the real test's 0.834 and reproduces the direction of the damage (13 clean-correct items break, 2 recover). It also reproduces the phonetic shape of the real errors, concentrated on emphatic and sibilant consonants once high-frequency energy is gone: *al-khayārāt* "the options" arrives as *al-sayyārāt* "the cars", and the reasoner receives fluent Arabic naming the wrong things. Error composition does not transfer, because dev labels are balanced and test labels are not (Table 1), but level and direction do.

Table 4 summarizes five transcript-level interventions aimed at the Arabic loss. Decoder hardening and genre priming buy about 2% relative WER, which cannot move 0.834. Scoring each option independently against the image changes nothing, ruling out position bias: the apparent drift toward option 0 among errors matches the label base rate. Marking low-confidence words for the reasoner applies as designed, with 26.6% of words marked at a threshold of 0.8 chosen from a mea-

| Intervention | Result | $p$ |
|---|---|---|
| Beam search + fallback | WER $0.244 \rightarrow 0.239$ | 0.077 |
| Arabic genre priming | WER $0.244 \rightarrow 0.238$ | 0.824 |
| Per-option scoring | acc $0.840 \rightarrow 0.830$ | 1.000 |
| ASR confidence marks | acc $0.840 \rightarrow 0.840$ | 1.000 |
| Ar+En dual decoding | acc $0.840 \rightarrow 0.850$ | 1.000 |
| Nothing (control) | 9/100 answers changed | – |

Table 4: Five interventions on the degraded-audio surrogate, against a control with nothing changed. Sign test for WER rows, McNemar for accuracy. The control's churn is pure API non-determinism, as Whisper's transcripts were byte-identical across runs.

sured reliability curve (words above it are right 90% of the time, below it 46%). It moves the predicted distribution from 42/35/23 to 38/36/26, close to the gold 38/35/27, yet accuracy does not change: matching the label distribution did not translate into correct answers. A second decoding of the same audio into English recovers individual items (*medicine* corrected to *brick*) yet is null in aggregate (0.856 to 0.856).

The control row changes how all of these results should be read. Rerunning the identical system on identical audio changed 9 of 100 answers, which is pure GPT-4o non-determinism at temperature 0. Every intervention's churn (12–16 items) sits barely above that floor, and at $n$=100 a true +0.05 gain would be detected 16% of the time, so these nulls rule out large effects only. Taken together, they still support one conclusion: the interventions downstream of the transcript have reached their ceiling, and the remaining Arabic loss occurs before any text exists. We recommend reporting a control run beside any LLM ablation and sizing dev sets to the effect sought (850 items for 80% power at +0.05).

## 6 Conclusion

We asked where the margins of our first-ranked (EN) and second-ranked (MSA) systems come from, and very little of the answer turned out to be model capacity. The largest English gain came from renaming the answer tokens, the Arabic gap was closed by replacing the listening stage, and the Task 2 ensemble won by breaking the rank ties that calibration creates. The post-submission study bounds what remains: the residual Arabic loss lies in acoustic modeling, and effects smaller than the measured non-determinism churn cannot be established without controls.

Three practical lessons carry beyond this shared task. First, before scaling a model or a prompt, it is worth checking whether the answer format itself is suppressing correct answers; a one-line change to the output vocabulary was worth more here than any model swap we tried. Second, when a benchmark offers parallel conditions that hold most inputs fixed, they are worth using: the paired tracks turned an accuracy gap into a diagnosis by isolating the single input that differed. Third, evaluations built on LLM judges should report a control run, because a judge that silently changes 9% of its answers between identical runs will otherwise convert noise into publishable-looking ablation differences.

## Limitations

Both dev splits are small (100 and 40 items), and the resulting variance is quantified rather than avoided: run-to-run SD is near 0.03 for Subtask 1a, and the bootstrap interval for Task 2's calibration step ($[-0.023, +0.055]$) spans zero even though the test set later landed the same change at +0.020. The degraded-audio surrogate matches the real test in level and direction but provably not in error composition, and its construction encodes our assumptions about recording conditions. The test label distribution in Table 1 is estimated from our own 0.962-accurate predictions, not from gold. All conclusions about annotator behavior rest on one annotation campaign in one cultural domain. The strongest configurations depend on closed commercial models, so we release code, prompts, per-item reasoning logs and the fully local variant (Whisper to Qwen, MSA dev 0.59) so the comparison is reproducible. The Sonnet-5 judge was never submitted alone, so the +0.074 is measured against the GPT-4o pipeline only.

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# A System details

Everything here is in the released code, but this section exists so the systems can be re-implemented from the paper alone.

## A.1 Subtask 1a cascade (submitted)

**Transcription** runs locally with faster-whisper large-v3, int8 compute, voice activity detection on, and the language forced to the track (`en` or `ar`), and the transcript is the concatenation of segment texts. **Reasoning** is one GPT-4o call at temperature 0 with a 600-token budget, three HTTP retries with linear backoff. The prompt tells the model the text is an automatic transcript of a spoken question followed by three options in spoken order, warns that recognition errors are likely, especially on names and cultural terms, and asks it to reconstruct the question and the three options before answering. The required JSON reply has four fields: the cleaned question in its original language, the three options in spoken order, one sentence of visual evidence describing only what is visible, and the answer as the word *first*, *second* or *third*. The prompt's one behavioral instruction: when two options seem plausible, choose the one the image evidence favours, never the one that merely sounds culturally typical.

## A.2 Answer parsing and fallback

The parser extracts the JSON object between the outermost braces, tolerating markdown fences,

and reads ordinal words before any digit. It never scans for bare digits: an earlier version did, and a digit inside a quoted option list matches as readily as an answer, which silently sent all 17 such test replies to option 0. A reply with no readable answer triggers one forced-choice retry (48 tokens: "answer with exactly one word"), and only a second failure falls back to option 0, the most frequent label. The fallback fired on 1 of 1,000 English test items and is flagged in the logs rather than silent. Runs append to disk after every item and resume by item id, with the system variant encoded in the output filename so no two systems can resume from each other's rows. An early filename collision between two variants produced a plausible-looking but meaningless 0.56, which is why the rule is mechanical.

### A.3 Subtask 1a joint baseline

Qwen2.5-Omni-3B in 4-bit NF4 with double quantization, float16 compute, greedy decoding, 24 thinker tokens (160 for a diagnostic variant that asks for a transcript first). A 1,000-item track takes about 1.4 hours on one consumer GPU. Audio and image go in a single request, and the prompt asks for the same JSON as the cascade plus a verbatim transcript field, which is what let us read *what the model heard* when diagnosing the MSA failure.

### A.4 Task 2 judges

Both judges run on GPT-4o through OpenRouter in JSON mode at temperature 0.4, three samples each, per-dimension median, with token budgets of 2,000 (checklist) and 3,000 (staged) that escalate only when the API reports truncation, capped at 4,000. The **checklist judge** lists 4–8 concrete cultural elements visible in the reference image ("white thobe with Qatari collar style", not "clothing"), gives a verdict per element, present-correct, present-flawed or absent, each with one line of visual evidence, and computes CEA as (correct + 0.5·flawed)/total, minus at most 0.1 for severe flaws. The **staged judge** must record, in order: a description of the generated image, then the elements the caption requires (each typed as attire, architecture, object, practice or setting), then a verdict and evidence line per element, then a list of fabricated elements (an empty list is stated to be the normal outcome), and finally authenticity notes comparing the scene against the reference. Only then does it score CC, CS, CI and HP, each defined in terms of its own recorded verdicts. Both prompts carry the training-annotation statistics verbatim (median composite 0.48, a quarter of instances at or below 0.15, mean CS 0.31, HP zero on 97% of instances) and the instruction that when torn between two scores, pick the lower. In code: CS is snapped to the five legal tiers, and HP is set to 0.0 whenever the fabrication list is empty.

### A.5 Calibration and ensemble

Isotonic regression is fitted per dimension on the 120 training instances and applied to CEA, CC, CI and HP, while CS is left alone because it is tier-valued, is our strongest dimension, and remapping would push it off the legal tiers. The ensemble averages the GPT-4o and Claude Sonnet 5 score tables per dimension with equal weights, snaps CS back to the nearest tier, and recomputes the composite from the official formula so the submission passes the organizer's consistency check.

### A.6 External tools and models

Beyond the released task data: Whisper large-v3 via faster-whisper, Qwen2.5-Omni-3B via HuggingFace transformers with bitsandbytes NF4 quantization, ffmpeg for the post-submission audio degradation, and the GPT-4o and Claude Sonnet 5 APIs through OpenRouter. No external datasets, no fine-tuning, and no data beyond the organizers' releases were used in any submitted run.

### A.7 Cost and runtime

The cascade costs about $5 per 1,000 items in API charges (transcription is local and free), and the joint system costs no API charges at all. Task 2 costs six calls per instance per model family, plus a one-time 720 calls to score the training split for calibration. Every run writes a per-item evidence log (transcripts, verdicts, raw replies), which is what every error analysis in this paper is computed from.

## B Verbatim prompts

Figures 3 to 7 reproduce every prompt exactly as sent to the models; only line breaks are adjusted to the page width. The cascade prompt receives the Whisper transcript appended after its final line, and both Task 2 prompts receive the images as attachments alongside the text.

```
You are given an image and an automatic transcript of a spoken multiple-choice
question about it. The transcript may be in Modern Standard Arabic or English,
and it contains a question followed by three answer options in the order they
were spoken.

Speech recognition makes mistakes, especially on names and cultural terms.
Reconstruct the intended question and the three options as faithfully as you
can; if the transcript is garbled, rely on the image and on which reading is
plausible.

Return JSON:
1. question: the question, cleaned up (keep it in its original language).
2. options: the three options as a list, in spoken order.
3. evidence: one sentence on what the image actually shows that decides the
answer. Describe only what is visible.
4. answer: "first", "second", or "third" - the option the image supports.

If two options seem plausible, pick the one the image evidence favours, never
the one that merely sounds culturally typical.

Return ONLY: {"question": "...", "options": ["...", "...", "..."], "evidence":
"...", "answer": "first"}

Transcript:
```

Figure 3: Subtask 1a cascade prompt (submitted system); the transcript is appended after the final line.

```
You will hear a spoken multiple-choice question about the image shown, followed
by three spoken answer options in order. The first option spoken is 0, the
second is 1, the third is 2.

Work through this and return JSON:
1. transcript: what you heard, verbatim.
2. question: the question alone.
3. options: the three options as a list, in spoken order.
4. evidence: one sentence on what the image actually shows that decides the
answer. Describe only what is visible.
5. answer: the index (0, 1, or 2) of the option that the image supports.

If two options seem plausible, pick the one the image evidence favours, never
the one that merely sounds culturally typical.

Return ONLY: {"transcript": "...", "question": "...", "options": ["...", "...",
"..."], "evidence": "...", "answer": 0}
```

Figure 4: Subtask 1a joint-system prompt (end-to-end baseline).

```
Answer with exactly one word and nothing else: first, second, or third.
Which of the three spoken options does the image support?
```

Figure 5: Subtask 1a retry prompt, used once when the first reply cannot be parsed.

```
You are an expert evaluator of cultural element accuracy in AI-generated images
of Qatari cultural scenes, calibrated to match expert human annotators from the
region.

You will see a REFERENCE image (an authentic Qatari cultural scene - this is the
ground truth) and a GENERATED image produced from a caption. Your single job:
measure how well the generated image reproduces the cultural elements of the
reference scene.

CALIBRATION (from 120 human-annotated instances): human CEA scores span the full
range - mean 0.55, a quarter of images at 0.20 or below, a quarter at 1.00.
Generations from generic captions usually LACK the reference's cultural elements
and score 0.0-0.3. Do not hesitate to give 0.0 or 1.0.

Protocol, recorded in the JSON:

STEP 1 - reference_elements: list the 4-8 core cultural elements VISIBLE IN THE
REFERENCE IMAGE that define the scene (specific attire like
thobe/ghutra/agal/abaya/batoola, architectural features, culturally significant
objects, setting). Concrete and checkable, not generic ("white thobe with Qatari
collar style", not "clothing").

STEP 2 - element_check: for each reference element, is the SAME TYPE of element
present in the GENERATED image?
  "present-correct" = present and culturally right (same type counts;
pixel-level similarity NOT required)
  "present-flawed" = present but culturally wrong in style/region/construction
  "absent" = not there
Give one line of visual evidence per verdict.

STEP 3 - CRAI_CEA in [0,1]: (count of present-correct + 0.5 * count of
present-flawed) / total elements, then adjust down by up to 0.1 if the flaws are
severe cultural errors. Round to two decimals.

Return ONLY JSON:
{"reference_elements": ["..."], "element_check": [{"element": "...", "verdict":
"...", "evidence": "..."}], "CRAI_CEA": 0.0}
```

Figure 6: Task 2 checklist-judge prompt (CEA).

```
You are an expert cultural accuracy evaluator for AI-generated images of Qatari
and Arab cultural scenes. Your scores must match expert human annotators from
the region, who are far stricter than typical AI judges.

CALIBRATION FACTS (measured on 120 human-annotated training instances):
- Median human composite score is 0.48; a quarter of images score below 0.15.
Low scores are normal.
- Human CS (cultural specificity) mean is 0.31: most generated images are NOT
culturally specific.
- Humans assign HP=0.0 on 97% of images: hallucination penalties are RARE,
reserved for clear fabrications.
- AI judges systematically score 0.2-0.3 higher than humans. Correct for this:
when torn between two scores, pick the lower.

Follow this exact protocol and record every stage in the JSON.

STAGE 1 - INDEPENDENT PERCEPTION
1a. generated_description: 2-4 sentences on what is ACTUALLY VISIBLE in the
generated image: people, attire (name specific garments), architecture, objects,
setting, style. Only what you can see.
1b. expected_elements: from the CAPTION (using the reference image as visual
ground truth for how each element should look), list every cultural element a
faithful generation must contain. Type each one: attire | architecture | object
| practice | setting. If the caption is generic (no cultural terms), list only
what the caption literally requires.

STAGE 2 - GROUNDED VERIFICATION
2a. element_check: for EVERY expected element: verdict "present-correct",
"present-flawed" (present but wrong style/region/construction, e.g. a turban
where ghutra+agal is expected, Moroccan arch where Qatari architecture is
expected), or "absent" - each with one line of visual evidence.
2b. fabricated_elements: culturally significant elements visible in the
generated image that are BOTH (i) not required by the caption AND (ii)
culturally wrong or misleading in this context. Extra but culturally plausible
detail is NOT fabrication. An EMPTY list is the normal outcome.
2c. authenticity_notes: 1-2 sentences comparing with the reference image: does
this look like real Qatar (materials, garment construction, colors, environment)
or like a generic stereotyped "Middle East"?

STAGE 3 - MECHANICAL SCORING (derive from your Stage 2 verdicts; do not re-judge
from overall impression)
CRAI_CEA [0-1]: fraction of expected elements "present-correct", counting
"present-flawed" as half credit; subtract up to 0.2 for severe flaws. All
correct = 0.9-1.0; about half = 0.4-0.6; mostly absent/wrong = 0.0-0.3.
CRAI_CC [0-1]: are depicted elements combined into a culturally plausible scene
(setting, pairings, context)? Fully plausible = 0.8-1.0; noticeable incongruity
= 0.4-0.7; incoherent = 0.0-0.3.
CRAI_CS - EXACTLY one of 0.0, 0.25, 0.5, 0.75, 1.0, judged on the GENERATED
IMAGE ALONE (ignore what the caption asked for):
  1.0 = unmistakably Qatari (e.g. batoola, Qatari landmark, Qatari-specific
garment construction)
  0.75 = clearly Gulf/Khaleeji but not uniquely Qatari
  0.5 = clearly Arab-world, not Gulf-specific
  0.25 = vague generic "Middle Eastern" flavor
  0.0 = no cultural markers at all.
  Most generated images fall in 0.0-0.5.
CRAI_CI [0-1]: truthfulness and respect. Costume-like inaccuracy, stereotyping,
exoticization, or distortion of real practices = below 0.5. Accurate and
dignified = 0.8-1.0.
CRAI_HP [0-1]: 0.0 if fabricated_elements is EMPTY (the usual case). 0.1-0.3 =
one minor fabrication; 0.4-0.7 = several or one severe; 0.8-1.0 = pervasive.
NEVER assign HP > 0 without a named element in fabricated_elements.

Return ONLY valid JSON:
{"generated_description": "...",
 "expected_elements": [{"element": "...", "type": "..."}],
 "element_check": [{"element": "...", "verdict": "...", "evidence": "..."}],
 "fabricated_elements": [],
 "authenticity_notes": "...",
 "CRAI_CEA": 0.0, "CRAI_CC": 0.0, "CRAI_CS": 0.0, "CRAI_CI": 0.0, "CRAI_HP":
0.0}
```

Figure 7: Task 2 staged-judge prompt (CC, CS, CI, HP).